ALLIANCE UNIVERSITY | NAAC GRADE A+ ACCREDITED UNIVERSITY

Alliance School of Advanced Computing

Capstone Project – I [7CS1999] [Batch : 2023 – 2027]

Project Title:

MedFedAI: Federated Intelligence for Privacy-Preserving Medical Image Classification

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I. Project Overview

- **Problem Context:** Problems associated with medical image classification include limited training data, class imbalance, resemblance of the diseases, and variance in the data sets. Consequently, CNNs, Transformers, transfer learning, and hybrid methods have been proposed as potential solutions.

- **Motivation:** Manual analysis of medical images is time-consuming and requires expert knowledge. The research aims to develop accurate and robust AI models that can learn meaningful features from medical images.

- **Significance:** Automated medical image analysis can help reduce diagnostic workload and support faster decision-making. It can also improve the consistency and accuracy of disease classification.

- **Application Domain:** The research focuses on AI-based medical image classification and disease detection. It can be applied to X-ray, CT, ultrasound, retinal, skin, and microscopy images.

- **Target Users:** This system is designed mainly for radiologists, medical practitioners, and medical researchers. The system can also serve AI researchers and healthcare organizations that require medical image analysis.

II. Literature Review

S.NO	Author Name Publication Year	Research / Methodology	Performance Metrics	Advantages	Limitations
1	Pan et al., 2026	**MedCTM: A CNN-Transformer-Mamba Hybrid Network for Medical Image Classification**. Combines CNN, Transformer and Mamba branches for local features, long-range dependencies and feature interaction.	Evaluated on **8 medical image datasets** covering 7 modalities and 100,000+ samples. Achieved **1.42–2.63 GFLOPs** and improved average F1 by **0.5–1.7%** over MedViTv2 while using **12.9–24.4% fewer parameters**.	Strong combination of local and global feature learning; better accuracy–efficiency trade-off; relatively lightweight.	More complex architecture because it combines three modelling paradigms; extensive multi-dataset validation is still needed for clinical deployment.
2	Si, Zhang & Zhao, 2026	**A Medical Image Classification Algorithm Based on a Hierarchical and Complementary Attention-Enhanced Swin Transformer Model**. Uses multi-scale, global-context and local-detail enhancement with Swin Transformer.	Tested on **8 MedMNIST v2 subsets**; significant improvements reported on **BreastMNIST, OCTMNIST, OrganSMNIST and PneumoniaMNIST**, with strong robustness and generalization across modalities.	Captures multi-scale lesions and subtle anatomical features; strong cross-modality evaluation.	Still based on Transformer architecture, so computational requirements can remain high; results are mainly benchmark-based rather than clinical validation.
3	Ren et al., 2025	**Conv-SdMLPMixer: A Hybrid Medical Image Classification Network Based on Multi-Branch CNN and Multi-Scale Multi-Dimensional MLP**. Combines multi-branch CNN with a multi-scale MLP feature-fusion module	On **BUSI**, achieved **93.33% accuracy, 93.65% F1, 94.78% precision and 92.66% recall**. Also evaluated on COVID19-CT, Small-ISIC2018 and Chest-Xray.	Captures local lesion details and broader spatial/channel information; performs well on small-lesion and noisy datasets.	Hybrid design increases architectural complexity; performance may depend on dataset characteristics and requires broader external validation.
4	MedSetFeat++ authors, 2026	**MedSetFeat++: An Attention-Enriched Set Feature Framework for Few-Shot Medical Image Classification**. Uses multi-head/multi-scale attention, positional embeddings, adaptive head gating and CBAM.	Under **2-way 10-shot 15-query**, achieved **92.17% accuracy on HAM10000, 70.89% on BreakHis and 73.46% on Kvasir**.	Useful for **limited labelled data**; strong few-shot performance; combines spatial, semantic and contextual features.	Few-shot results are task/setup dependent; evaluation is limited to selected datasets and episode settings.
5	Asimeng et al., 2026	**Transformer-Masked Autoencoder (MAE) for Robust Medical Image Classification: A Comprehensive Survey**. Reviews MAE-based approaches for X-ray, MRI, CT, ultrasound and histopathology.	Survey of **110 studies**; reports that MAE initialization improves label efficiency, calibration and cross-scanner transfer, especially with parameter-efficient fine-tuning.	Very useful for identifying current trends in **self-supervised learning and label-efficient classification**.	It is a **survey**, not a new classification model; reported studies use different datasets and evaluation protocols, making direct comparison difficult.

II. Literature Review

S.NO	Author Name Publication Year	Research / Methodology	Performance Metrics	Advantages	Limitations
6	Zheng et al., 2024	Proposed **MedPrompt**, a weakly supervised prompt-learning approach for low-resource medical image classification.	MedPrompt-ViT achieved **AUC 0.658 (CheXpert), 0.616 (MIMIC-CXR), 0.9553 (COVID), 0.8304 (RSNA)** in full-shot evaluation.	Reduces dependence on manually designed expert prompts; supports zero-shot and few-shot classification.	Prompt wording can strongly affect performance; domain-expert dependence is reduced but not completely eliminated.
7	Liu et al., 2024	Proposed **Eff-CTNet**, an efficient alternating CNN–Transformer architecture with Group Cascade Attention.	On COVID19-CT: **92.57% accuracy, 93.17% F1, 91.46% precision, 94.94% recall, AUC 0.9240**.	Efficiently combines local and global features while reducing attention redundancy and computational cost.	Slight overfitting caused by noise and class imbalance was observed on Chaoyang.
8	Wu et al., 2023	Proposed **CTransCNN**, combining CNN and Transformer branches for multilabel medical image classification.	Average AUC **83.37%** on ChestX-ray11 and **78.47%** on NIH ChestX-ray14; AUC **84.56%** and mAP **67.67%** on TCMTD.	Models local/global features, label correlations and class imbalance jointly.	Severe label imbalance; some datasets have limited samples and the hybrid architecture increases computational complexity.
9	Manzari et al., 2023	Proposed **MedViT**, a hybrid CNN–Vision Transformer for generalized medical image classification.	Average AUC: **93.6%, 94.2%, 93.5%** and average accuracy: **84.0%, 85.1%, 84.2%** for MedViT-T, S and L.	Combines CNN local features with Transformer global features; good accuracy–complexity trade-off.	Transformers require substantial training data and computation; medical images have large texture, shape and size variations.
10	Wang et al. / MedFMC, 2023	Introduced **MedFMC**, a benchmark for foundation-model adaptation and few-shot medical image classification across multiple modalities.	Fine-tuned DenseNet-121, EfficientNet-B4 and Swin-Base; performance evaluated using **Accuracy, AUC and mAP** across five datasets.	Supports few-shot and transfer-learning evaluation across diverse medical tasks and modalities.	Performance drops with very few patient samples; head–tail class imbalance remains a challenge.

II. Literature Review

S.NO	Author Name Publication Year	Research / Methodology	Performance Metrics	Advantages	Limitations
11	Kim et al., 2022	Conducted a literature review of **transfer learning with pretrained CNNs** for medical image classification.	Reviewed **425 articles**, with **121 studies** meeting the inclusion criteria; Inception was the most frequently used deep model.	Shows transfer learning is effective with limited medical data; provides recommendations for selecting CNN backbones and TL strategies.	Focused mainly on pretrained CNNs; Vision Transformers and source–target domain differences were not evaluated
12	Huynh, Nibali & He, 2021	Proposed **Adaptive Blended Consistency Loss (ABCL)** for semi-supervised medical image classification with class imbalance.	On HAM10000, UDA-WeightedCE-ABCL achieved **UAR 0.68, G-mean 0.66, average AUC 0.96**.	Handles both limited labels and class imbalance; improves minority-class performance.	Severe class imbalance remains challenging; strong augmentation can reduce the effectiveness of ABCL.
13	Azizi et al., 2021	Studied **self-supervised pretraining** with SimCLR and proposed **MICLe** for medical image classification.	**+6.7% top-1 accuracy** in dermatology and **+1.1% mean AUC** in chest X-ray classification.	Reduces dependence on labelled data; improves robustness to distribution shift and label efficiency.	Requires suitable large-scale unlabeled data and substantial computational resources; domain differences between natural and medical images remain important.
14	Yadav & Jadhav, 2019	Compared **ORB+SVM, VGG16 transfer learning, InceptionV3 transfer learning and Capsule Network** for pneumonia classification from chest X-rays.	Transfer learning was the best of the three main approaches; VGG16 with one retrained convolution layer gave the best result.	Effective for small datasets; data augmentation improves performance.	Fine-tuning can overfit; model complexity must match dataset size; visualization and stronger CNNs were suggested as future work.
15	Smitha Rajagopal et al., 2025	FL + Deep CapsNet for brain-tumor MRI classification with U-Net feature extraction.	96.79% Accuracy, 96.84% Recall, 96.85% F1, 0.97 AUC	Privacy-preserving, decentralized training, good classification performance.	High computational cost, limited scalability, Non-IID handling needs improvement.

III. Research Gap

1. Identified Research Gap:

Federated learning allows healthcare institutions to train a shared model while keeping medical images at their own sites. However, sharing model updates may still expose information about the images used for training. The current MedFedAI plan does not yet define a specific privacy threat model, a method to test information leakage, or a protection mechanism with measurable privacy settings.

2. Technical Challenges:

- Shared model updates may reveal sensitive information about local medical images.
- The project must define what information a potential attacker can access.
- Privacy leakage needs to be assessed using a consistent and repeatable test.
- A privacy mechanism must be configured and its privacy settings recorded.
- The Deep Capsule Network and federated training process must operate without transferring raw images.
- Privacy testing must fit within the available computing resources.

3. Need for Proposed Work:

MedFedAI will combine a Deep Capsule Network with federated learning for medical image classification. The project will focus on testing whether shared model updates expose private information and evaluating a privacy-protection mechanism. The work will report the privacy test results, the protection settings, and the limitations of the proposed framework.

IV. Requirements Analysis

Software Requirements

1. Python → project implementation
2. PyTorch or TensorFlow → Deep Capsule Network development
3. NumPy and Pandas → image data organization and processing
4. Scikit-learn → data preparation and dataset splitting
5. Flower or TensorFlow Federated → federated learning simulation and FedAvg
6. Differential-privacy library compatible with the selected framework → update protection and privacy-budget tracking
7. Matplotlib → visualization of privacy evaluation results
8. Google Colab or Jupyter Notebook → experiments
9. Git and GitHub → version control

Dataset / Resource Requirements

- Labelled medical image datasets for classification
- Simulated client environments representing healthcare institutions
- Separate local data storage for each client
- Computing resources, such as GPU or CPU, memory, and storage
- Access to shared model updates for controlled privacy testing
- Experiment logs to record privacy settings and evaluation results

IV. Requirements Analysis

Constraints:

1. Medical images contain sensitive information and must remain at their respective client sites.
2. Shared model updates may expose information about local training images.
3. Privacy protection requires careful selection of update clipping, noise settings, and privacy-budget parameters.
4. Limited computing resources may restrict the number of clients and privacy-attack experiments.
5. Communication between clients and the server may increase training time.

Proposed Differentiation:

MedFedAI combines Federated Learning with a Deep Capsule Network for medical image classification and makes privacy the central focus. Raw images remain at each client while model updates are shared for aggregation. The project will assess whether those updates expose information and evaluate a privacy-protection mechanism, such as differential privacy, to reduce potential leakage.

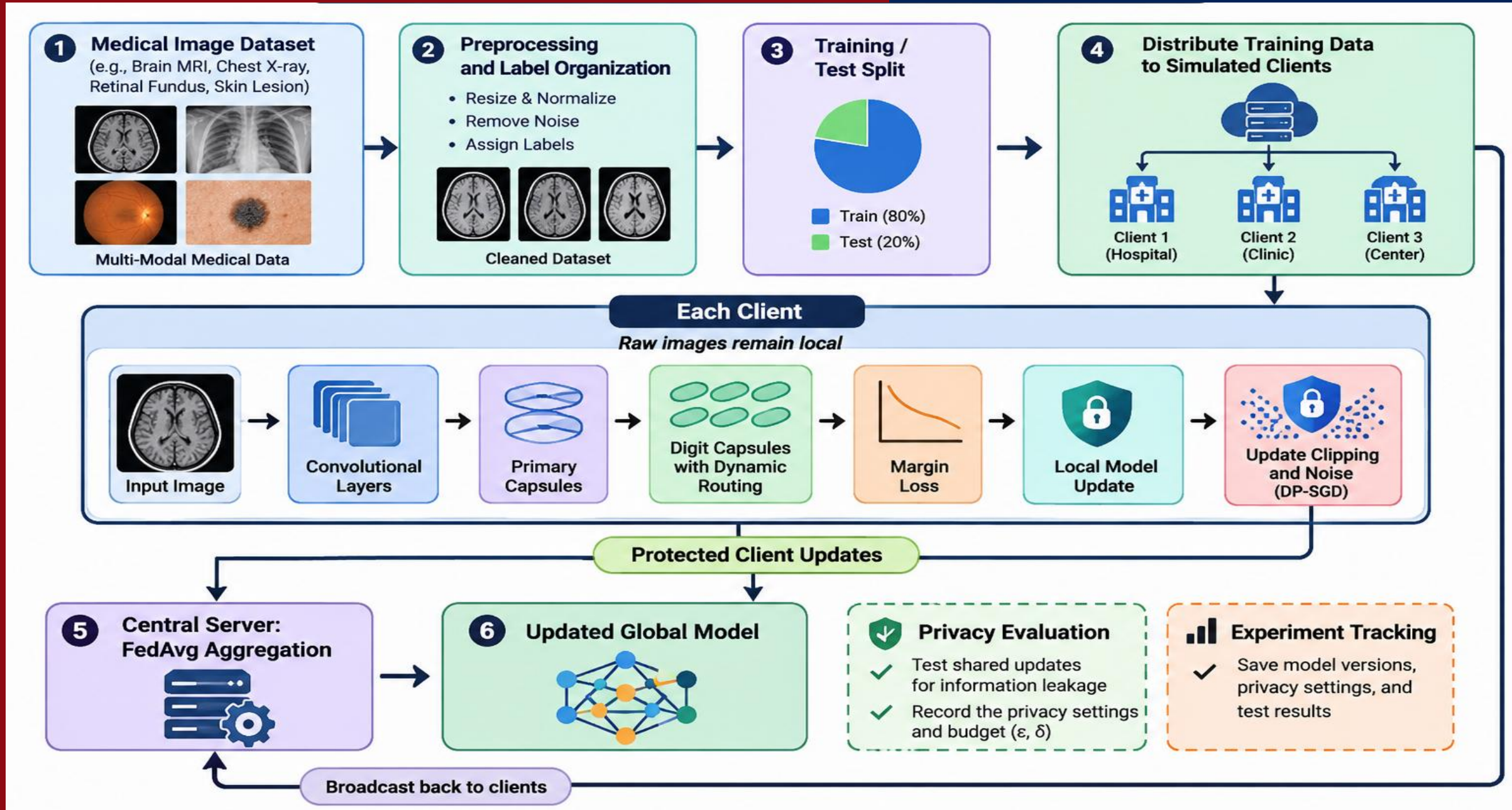
V. Problem Statement

Medical images contain **sensitive patient information**, so sharing them directly between healthcare institutions can create privacy risks. **MedFedAI** uses federated learning with a Deep Capsule Network for medical image classification while keeping raw images at each client. However, **shared model updates may still reveal information about the images used during training**. This project will evaluate **privacy risks** in the federated training process and investigate client-level differential privacy to protect the shared updates.

VI. Objectives

1. **Develop a privacy-preserving federated learning framework** using a Deep Capsule Network (Deep CapsNet) for medical image classification, with each client keeping its images locally and sharing only model updates.
2. **Evaluate privacy risks in shared model updates** by defining an attacker's access and testing whether a controlled reconstruction or inference attack can recover information from those updates.
3. **Apply and assess a privacy-protection mechanism** such as differential privacy, then compare the protected system with the baseline and report the privacy results and limitations.

VII. Architecture



VIII. Proposed Methodology stepwise

1. **Dataset Acquisition:** Select labelled medical images for the classification task and organize them into disease or condition classes.
2. **Preprocessing:** Resize images to a consistent dimension, normalize pixel values, and apply suitable augmentation to prepare them for model training.
3. **Client Distribution:** Separate training and testing data, then distribute the training images among simulated healthcare clients. Each client keeps its images locally.
4. **Local Model Training:** Train a Deep Capsule Network independently at each client using convolutional layers, primary capsules, dynamic routing, and margin loss.
5. **Federated Aggregation:** Use Federated Averaging (FedAvg) to combine client model updates at the central server without transferring raw medical images.
6. **Privacy Risk Assessment:** Define what the server can observe and test whether an attacker can infer or reconstruct information from the shared model updates.
7. **Privacy Protection:** Apply differential privacy to the updates using clipping and calibrated noise. Record the privacy settings and budget.
8. **Privacy Evaluation:** Compare the information recovered from updates before and after protection. Report the privacy findings, the stated guarantee, and the method's limitations.

IX. Implementation Results

1. Implementation :

https://colab.research.google.com/drive/1Ei5QddO2s0S1gP702YYozlw3__ZABXhU

2. Results:

Parameter	Chest X-ray Pneumonia	Brain Tumor MRI	Digital Knee X-ray
Model	Federated Deep CapsNet	Federated Deep CapsNet	Federated Deep CapsNet
Image Size	128 × 128	128 × 128	128 × 128
Classes	2	4	2
Clients	4	4	4
Fed. Rounds	3	3	3
Local Epochs	1	1	1
Batch Size	4	4	4
Learning Rate	0.0005	0.0005	0.0005
Training Images	1,000	2,000	800
Testing Images	624	1,600	200
Accuracy (%)	67.95	56.06	46.50
Precision (%)	74.13	60.76	44.63
Recall (%)	67.95	56.06	46.50
F1 Score (%)	60.30	51.43	41.40
Test Loss	0.2081	0.2645	0.2773

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Thank you

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